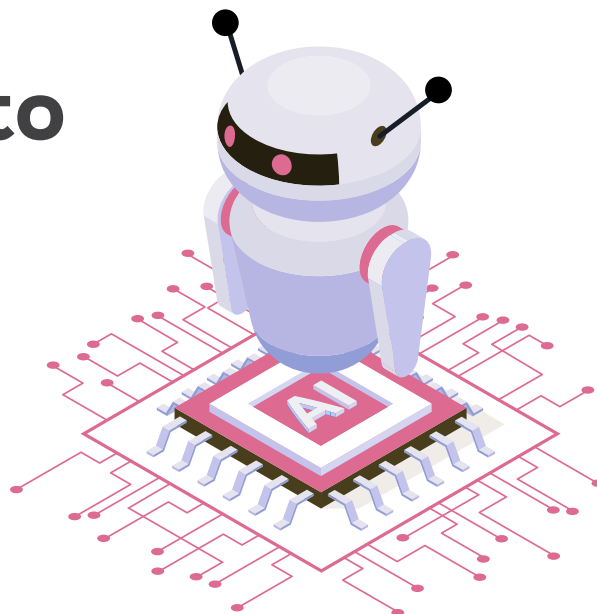


A Deep Dive into GPT-3

Generative AI, particularly OpenAI's GPT-3, has revolutionised the landscape of artificial intelligence and natural language processing. Here we explore the capabilities of GPT-3 and the broader implications of open source generative AI.



Generative AI is very exciting technology. It's like having a robot apprentice who can learn how to make art, music, and more! However, it still takes a lot of data, computing power, and learning for AI to match the creative abilities of people. But generative AI is getting better every day as it studies more examples of human creativity. The most amazing thing is that AI is not just copying work made by humans. It is able to take what it has learned and come up with new patterns, combinations, and innovations. Generative AI will help make art and content that has never been seen before!

Open source generative AI

The code used to create open source generative AI is available for everyone to look at, change, and use to build new things. It is not kept a secret by one company. Open source is about sharing and working together. Some examples of open source generative AI projects are:

- **RunwayML:** This helps people create new drawings, sounds, and videos with AI. You can change the code to customise the AI art!
- **StyleGAN:** This AI makes realistic fake human faces. The code is public so new things can be generated.
- **Magenta:** A tool from Google to make music with AI. You can build

on it to train the AI in new ways.

- **TensorFlow:** This popular programming library lets anyone build all kinds of AI models. Generative AIs are created with it. Some other cool open source projects are GAN Lab, Mimetic, Lexica, and VQGAN. The code for all these is free to use and change! The benefit of open source is that anyone can learn from these projects, make them better, and create new innovations. You may be asking yourself, "Why should I choose open source generative AI over closed source?" The key advantage of open source generative AI is the collaborative and transparent nature of its development. With the code and models being public, many more developers can contribute, spot issues, suggest improvements, and build custom solutions suited to their needs. This pooling of knowledge and effort can often outpace what even large companies can do internally.

Understanding GPT-3

Have you ever wondered how your phone is able to predict the next word as you type? Or how can a website automatically translate into different languages? Behind the scenes, these apps use very complex artificial intelligence systems called neural networks. GPT-3 is one of the newest and most advanced

neural networks ever made! GPT-3, or the Generative Pre-trained Transformer 3, is like a super-smart computer that is really good at understanding and generating human-like text. Created by OpenAI, think of it as a digital friend that can chat with you, write stories, answer questions, and do many other cool things with words.

A neural network is a type of AI algorithm designed to recognise patterns. It is inspired by the human brain. The 'neurons' are maths functions that connect in complex ways, similar to networks of real neurons in our heads. By analysing tons of data, the neural network learns to make predictions and decisions without needing explicit programming. This is the same way OpenAI's GPT-3 works.

GPT-3 has been trained on a massive amount of text from the internet. It's like having read billions and billions of books, articles, and conversations. This training allows it to understand the patterns and styles of human language. Now, when you chat with GPT-3, it uses its knowledge to generate responses. You can ask it about history, science, or even tell it to create a story, and it will do its best to come up with interesting and relevant answers. It was trained on nearly half a trillion words from web pages, books, Wikipedia and all kinds